Emissions Inventories: OWID / Global Carbon Project and PRIMAP-hist

> Part of the Functional Series. This document describes the two sources the platform uses for **emissions** — who produces each, what it counts, how far it reaches, and why there are two. Concentration and temperature are in the next document.

1. Two sources, two jobs

	OWID / Global Carbon Project	PRIMAP-hist
Producer	Our World in Data republishing the Global Carbon Project's Global Carbon Budget data	The PRIMAP research group (Gütschow, Pflüger and colleagues), published on Zenodo
Gases	CO_2 only (fossil fuels + cement; land-use change as a separate column)	CO_2, CH_4, N_2O and F-gases, with the Kyoto-basket total
Used for	The country-level analysis: trends, forecasts, scenarios, the Overview, Country Profile and Scenario pages	Total greenhouse gases, gas composition and each country's share of the global total
Years	1750–2024	1750–2024 (v2.8; 2025 was published but excluded as incomplete)
Geography	254 entities in the file, of which 215 report CO_2 in 2024; the rest are regions, groupings and income classes	207 country and territory areas
Unit	Million tonnes of CO_2 ($MtCO_2$)	Million tonnes of CO_2-equivalent ($MtCO_2e$), AR5 GWP-100
Licence	CC BY 4.0 for OWID's compilation; Global Carbon Project data requires citing the original	CC BY-NC-SA 4.0: non-commercial use, attribution, share-alike

The split is deliberate. OWID's file is richer per country (population, GDP and per-capita values) but covers CO_2 only. PRIMAP-hist is the only source in the platform that puts the other gases on the same footing as CO_2, so it answers "what is the *total* and how is it composed?".

2. OWID / Global Carbon Project

What it measures. Annual CO_2 emissions from burning fossil fuels and making cement, by country, from the Global Carbon Project's annual Global Carbon Budget. Land-use-change CO_2 (deforestation, afforestation and similar) is a *separate* column, not part of the headline `co2` figure.

How it is produced. The Global Carbon Project combines energy statistics with emission factors for fossil CO_2, and averages several bookkeeping models for land-use change. OWID republishes the result as one tidy file, adds derived per-capita and cumulative columns, and updates it when a new Global Carbon Budget is released. The platform's refresh job re-downloads it monthly.

What to remember when reading it.

- **Fossil and land-use CO_2 differ in confidence.** Fossil CO_2 is comparatively well measured; land-use CO_2 is a modelled estimate with a large uncertainty (roughly ±2.6 $GtCO_2$ per year, one standard deviation, in recent decades, per the Global Carbon Budget). The platform keeps them separate and only adds them when it says so.
- **International aviation and shipping.** The World row includes them; the sum of the countries does not. The platform keeps both numbers (`co2_mt` and `national_sum_mt`) and uses the national sum as the denominator for country shares.
- **Not every row is a country.** The file mixes sovereign countries with continents, income groups, the European Union aggregates and special entries (for example, "Kuwaiti Oil Fires"). The platform filters these out of any country ranking using one list in the notebooks' shared constants, and its API does the same.
- **The year range depends on country.** A country's series begins when data begin: coverage in the 1990s differs for countries that did not yet exist.

3. PRIMAP-hist

What it measures. A single, consistent, long national series of greenhouse gas emissions since 1750 for CO_2, CH_4, N_2O and the fluorinated gases, covering the national total *excluding* land use, land-use change and forestry (category M.0.EL).

How it is produced. It is a composite. Country-reported data, where a country has submitted an inventory to the UN climate convention, take priority; gaps are filled from third-party datasets (including CDIAC, the Energy Institute, FAOSTAT and EDGAR) and harmonized into one series. The platform uses the *country-reported priority* scenario (HISTCR) from the file without extrapolation. The country-reported and third-party scenarios differ by about 4.7% in 2023, which is a good indication of the underlying uncertainty.

What to remember when reading it.

- **It is a reconstruction, not a measurement.** Values before about 1970 are rebuilt from historical datasets.
- **Explicit gaps, not zeros.** Methane and nitrous oxide cover 206 areas; F-gases cover about 150–170. An area without a gas is recorded as missing, never as zero.
- **No world aggregate and no international transport.** The world total is the sum of the reporting areas and so excludes international aviation and shipping.
- **A year is published only when it is complete.** The platform requires each gas to cover at least 98% of the previous year's emissions and the national total to be within ±15% of the prior year. The 2025 values in v2.8 fail this rule (methane coverage 2.0%, F-gases 0.0%, total –28% against 2024) and are held back rather than extrapolated.
- **Arithmetic is checked.** The platform recomputes the total from the gases (CO_2 + CH_4 × 28 + N_2O × 265 + the F-gas basket) and requires it to match PRIMAP-hist's own total within 0.5%. Across 1750–2024 the mismatch is below 0.5% in every year the platform publishes.

4. Why two sources disagree, and what the platform does about it

OWID/GCP and PRIMAP-hist report different 2024 CO_2 figures: 38,599 $MtCO_2$ (OWID, includes international transport) against 37,974 $MtCO_2$ (PRIMAP-hist, excludes it). Part of the gap is scope, since international aviation and shipping alone are about 1,192 $MtCO_2$ in OWID's 2024 figures. The remainder reflects different energy statistics, emission factors and treatment of gaps. The platform's position:

1. Use each source only for what it covers best, and **never silently mix them** in one chart.
2. Label each figure with its scope (gases, whether land use and international transport are included).
3. **Disclose** the difference rather than hide it by reconciling the two.

When the benchmark documents compare the platform with the IPCC and others, the same rule holds: the first question is always *which scope does this number use?*

5. The country sets

The platform works with three overlapping sets of countries:

Set	Size	How chosen	Used for
Featured	10	Hand-picked: China, United States, India, Russia, Japan, Germany, Brazil, United Kingdom, South Africa, Australia	Default charts and narrative
Expanded	40	Data-driven: at least 90% data coverage and at least 100 $MtCO_2$ in the latest year (2024). Together they hold 92.2% of the global total	Model training and evaluation, aggregate sums, tables and picker pools
All countries	~215	Every sovereign country with data	Overview rankings and the map

The expanded set is recomputed from live data and stored in `data/selected_countries.json`.

See also

- Concentrations and Temperature
- Data Quality and Harmonization
- Technical: Multi-Source Data Pipelines — how these sources are ingested, validated and published
- Technical: EDA and Data Engineering — how the OWID file is profiled and filtered